ENGINEERING CHECKLIST

Pre-Silicon Verification Sign-off Checklist

A practical review framework for assessing verification completeness, regression health, protocol checking and bug closure before design sign-off.

FOCUS AREAS

Coverage closure • Regression readiness • Assertions & protocol checks • Bug closure & sign-off criteria

WHO SHOULD USE THIS CHECKLIST?

Design Verification Managers, SoC Verification Leads, Verification Architects and engineering leaders preparing for pre-silicon sign-off.

HOW TO USE

Review each checkpoint against project evidence. Mark an item complete only when its supporting artifact, report, test result or approved waiver is available. Record the owner and evidence reference in the final column.

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This checklist is a planning aid. Project-specific sign-off criteria must be agreed by the design and verification teams and aligned with applicable IP, SoC and interface specifications.

Project sign-off record

Complete this record before reviewing the technical checkpoints. Use project-approved acceptance criteria; this checklist does not prescribe universal pass thresholds.

PROJECT / SoC		**IP / SUBSYSTEM**	
DESIGN REVISION		**SPECIFICATION REVISION**	
VERIFICATION OWNER		**REVIEW DATE**	
TARGET SIGN-OFF		**BUILD / REGRESSION ID**	

Recommended sign-off gates

GATE	EXIT EXPECTATION	STATUS
Coverage	Required functional coverage goals are met, waived or formally dispositioned.	■ Pass ■ Open ■ Waived
Regression	The agreed regression suite is stable, reproducible and free of unresolved blocking failures.	■ Pass ■ Open ■ Waived
Protocol	Applicable assertions, protocol checks and negative/error scenarios have been exercised and reviewed.	■ Pass ■ Open ■ Waived
Bug closure	Critical and high-severity issues are closed or have an approved disposition and documented risk.	■ Pass ■ Open ■ Waived
Approval	Design, DV and relevant integration stakeholders have reviewed the evidence and approved sign-off.	■ Pass ■ Open ■ Waived

Evidence pack to assemble

- ■ Verification plan and requirement traceability matrix
- ■ Coverage reports and waiver approvals
- ■ Regression dashboard, failure triage and rerun evidence
- ■ Assertion / protocol checker reports
- ■ Bug tracker export and unresolved-issue review
- ■ Final sign-off minutes and stakeholder approvals

01 | Coverage closure

Demonstrate that planned verification objectives have been exercised and remaining gaps are understood, justified and approved.

Mark each checkpoint only when supporting evidence is available.

OK	VERIFICATION CHECK	EVIDENCE / OWNER
	Verification plan maps each in-scope requirement and feature to verification objectives.	
	Functional coverage model is reviewed against specification and design intent.	
	Mandatory covergroups, coverpoints and crosses are implemented and enabled in sign-off configuration.	
	Coverage exclusions, illegal bins and unreachable scenarios have documented technical justification.	
	Coverage goals and closure thresholds are agreed and recorded.	
	Coverage is collected across approved regression tests and relevant configurations.	
	Uncovered bins are tested, justified as unreachable, or formally waived.	
	Critical feature combinations and corner cases are represented in functional coverage.	
	Code coverage, where used, is reviewed alongside functional coverage, not as a substitute.	
	Coverage database merge, filtering and reporting are repeatable and revision-correct.	
	Coverage waivers have an owner, rationale, impact assessment and approval.	
	Final coverage report is archived with source, seed, test and build identifiers.	

Reviewer prompts

Are high-risk requirements represented only by code coverage? Are critical crosses missing? Are exclusions hiding scenarios that should be tested?

02 | Regression readiness

Establish confidence that the approved regression suite is stable, repeatable and representative of the verification plan.

Mark each checkpoint only when supporting evidence is available.

OK	VERIFICATION CHECK	EVIDENCE / OWNER
	Sign-off regression suite is defined, version-controlled and approved by the verification owner.	
	Regression runs use approved RTL, testbench, VIP, configuration and tool versions.	
	Smoke, sanity, directed, constrained-random and negative tests are included as appropriate.	
	Tests cover required modes, configurations, reset sequences, power states and operating conditions.	
	Completion criteria, pass-rate expectations and stability thresholds are documented.	
	Tests have deterministic setup and reproducible seed / configuration capture where applicable.	
	Known failures are triaged; infrastructure, testbench and design failures are distinguished.	
	Flaky tests, intermittent failures and unexplained timeouts are resolved or dispositioned.	
	Failed tests are rerun with evidence that the original failure is understood.	
	Runtime, resource limits and parallel execution settings are reviewed for sign-off.	
	Nightly regression results show acceptable stability over the agreed observation window.	
	Final regression dashboard and run manifests are archived and linked to sign-off revision.	

Reviewer prompts

Is regression running against the exact sign-off RTL? Are failures understood? Is the final run reproducible?

03 | Assertions & protocol checks

Verify design intent and interface behavior with executable properties, protocol-aware checking and negative testing.

Mark each checkpoint only when supporting evidence is available.

OK	VERIFICATION CHECK	EVIDENCE / OWNER
	Assertions are mapped to key design requirements, interface rules and protocol invariants.	
	Assertions are enabled in sign-off configuration and activity / coverage is reviewed.	
	Reset, initialization, ordering, handshake, timeout and state transitions are checked where applicable.	
	Protocol checkers use the correct specification revision, mode and feature set.	
	Legal and illegal transaction sequences are exercised, including boundary conditions.	
	Error responses, malformed transactions, retry / replay and recovery paths are verified where applicable.	
	Backpressure, concurrency, outstanding transactions and ordering rules are checked.	
	Relevant clock-domain and reset-domain interactions are exercised.	
	Assertion failures are triaged; disabled or masked assertions have approved rationale.	
	VIP configuration, checker versions and protocol compliance settings are recorded.	
	Negative testing and error injection cover agreed fault scenarios.	
	Protocol checker and assertion results are archived with sign-off evidence.	

Reviewer prompts

Are key protocol rules asserted? Are checker configurations specification-aligned? Have negative and recovery scenarios been exercised?

04 | Bug closure & sign-off criteria

Ensure unresolved issues are visible, risk-assessed and formally accepted, and evidence supports the sign-off decision.

Mark each checkpoint only when supporting evidence is available.

OK	VERIFICATION CHECK	EVIDENCE / OWNER
	Bug tracker is baselined for sign-off, with severity and priority definitions agreed.	
	All critical and high-severity bugs are closed or have an explicitly approved exception.	
	Every open issue has documented impact, workaround (if any), owner and target disposition.	
	Fixes are verified with targeted tests and appropriate regression coverage.	
	Bug closure is linked to the relevant RTL / testbench revision and verification evidence.	
	Duplicate, rejected and not-a-bug issues have documented rationale and reviewer disposition.	
	Known limitations and residual risks are communicated to design, DV and integration stakeholders.	
	Waivers and accepted risks have named approvers and are stored with the sign-off record.	
	Requirements traceability, coverage, regression and bug reports use consistent design revisions.	
	Final sign-off review has approval from designated design and verification owners.	

Reviewer prompts

Have residual risks been accepted by named approvers? Are bug fixes tied to the correct RTL revision and regression evidence?

Final sign-off decision

DECISION	■ Approved ■ Approved with documented exceptions ■ Not approved
OPEN RISKS / WAIVERS	
EVIDENCE LOCATION	
DV OWNER / DATE	
DESIGN OWNER / DATE	
PROGRAM / INTEGRATION APPROVER	

NEED TO STRENGTHEN YOUR PRE-SILICON VERIFICATION SIGN-OFF?
Discuss your UVM architecture, functional coverage closure, regression readiness and protocol verification requirements with AionSi's engineering team.

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General engineering aid only. Adapt all gates and acceptance thresholds to the project verification plan, applicable interface specifications, safety requirements and customer sign-off process.